

TEN



LOSER TAKES ALL - ON THE NONLINEARITIES OF LIFE

The nonlinear viciousness of life. Moving to Bel Air and acquiring the vices of the rich and famous. Why Microsoft's Bill Gates may not be the best in his business (but please do not inform him of such a fact). Depriving donkeys of food.

Next I put the platitude that *life is unfair* under some examination, but in a new angle. The twist: life is unfair in a *nonlinear* way. This chapter is about how a small advantage in life can translate into a highly disproportionate payoff, or, more viciously, how no advantage at all, but a very, very small help from randomness, can lead to a bonanza.

The Sandpile Effect

First we define nonlinearity. There are many ways to present it, but one of the most popular ones in science is what is called the sandpile effect, which I can illustrate as follows. I am currently sitting on a beach in

Copacabana, in Rio de Janeiro, attempting to do nothing strenuous, away from anything to read and write (unsuccessfully, of course, as I am mentally writing these lines). I am playing with plastic beach toys borrowed from a child, trying to build an edifice – modestly but doggedly attempting to emulate the tower of Babel. I continuously add sand to the top, slowly raising the entire structure. My Babylonian relatives thought they could thus reach the heavens. I have more humble designs; to test how high I could go before it would topple. I keep adding sand, testing to see how the structure would ultimately collapse. Unused to seeing adults build sandcastles, a child looks at me with amazement.

In time – and much to the onlooking child's delight – my castle inevitably topples to rejoin the rest of the sand on the beach. It could be said that the last grain of sand is responsible for the destruction of the entire structure. What we are witnessing here is a nonlinear effect resulting from a linear force exerted on an object. A very small additional input, here the grain of sand, caused a disproportionate result, namely the destruction of my starter tower of Babel. Popular wisdom has integrated many such phenomena, as witnessed by such expressions as “the straw that broke the camel's back” or “the drop that caused the water to spill”.

These nonlinear dynamics have a bookstore name; chaos theory, which is a misnomer because it has nothing to do with chaos. Chaos theory concerns itself primarily with functions in which a small input can lead to a disproportionate response. Population models, for instance, can lead to a path of explosive growth, or extinction of a species, depending on a very small difference in the population at a starting point in time. Another popular scientific analogy is the weather, where it has been shown that a simple butterfly fluttering its wings in India can cause a hurricane in New York. But the classics have their share to offer as well: Pascal (the same of Pascal's wager in Chapter 7) said that if Cleopatra's nose had been slightly shorter, the world's fate would have changed. Cleopatra had comely features dominated by a thin and elongated nose that made Julius Caesar and his successor Marc Antony fall for her (here the intellectual snob in me cannot resist dissenting against conventional wisdom; Plutarch claimed that it was Cleopatra's skills in conversation, rather than her good looks, that caused the maddening infatuation of the shakers and movers of her day; I truly believe it).